<u>Set- 3</u>

HSS102: Concepts in Humanities and Social Sciences

April 2026

1. Explain using a diagram the efficient and inefficient outcomes within an economy with the help of the production possibility frontier. (Chapter 2, Principle of Microeconomics by Mankiw)
2. Explain the law of demand. Sketch a linear demand curve by assigning different values of quantities and prices of goods or services you are assuming. (Section 4-2, Chapter 4, Principle of Microeconomics by Mankiw)
3. Derive the market demand curve. Section 4-2b, Chapter 4, Principle of Microeconomics by Mankiw)
4. Describe the factors (internal and external) which lead to a shift in the demand curve. Section 4-2c, Chapter 4, Principle of Microeconomics by Mankiw)
5. What is the difference between quantity demanded and quantity demand? Show diagrammatically. (Section 2.1, Chapter 2, Microeconomics by Pindyck and Rubinfeld)
6. The price of biscuits increases from $10 to $12, and the quantity demanded decreases from 100 units to 80 units. Calculate the price elasticity of demand.

   Answer- |PED| = 1
7. A product's price decreases from $50 to $45, and quantity demanded increases from 200 to 250 units. Find the price elasticity of demand.

   Answer- |PED| = 2.5
8. Consider Apple as a normal product. Suppose the price elasticity of demand for apples is equal to .75. If the price of apples rises by 10%, we expect a ___________ decrease in apple consumption.

   Answer- % change in the quantity demanded = 7.5%
9. Jason is a seller of Klonks. He has seen the price of a Klonk rise from $10 to $14, and he has responded by increasing his daily production from 100 units to 140 units. Compute the arc elasticity of supply.

   Answer- Es = 1
10. Suppose purchases of notebooks increase by 3% when income increases by 5%. Given this, the income elasticity (EI) for notebooks is equal to?

    Answer- Ei = 0.6

11. If the income elasticity of demand for coffee is .40, then a 6% increase in consumer income would cause consumption of coffee to __________ by ___________. Coffee is a normal good.

Answer- 2.4%

12. Given: $Qs = -5 + 3P$; $Qd = 10 – 2P$. Find the equilibrium price and quantity.

Answer- $P^* = 3$; $Q^* = 4$

13. Given the demand function

$$P + Q^2 + 3Q – 20 = 0$$

And the supply function

$$P – 3Q^2 + 10Q = 5$$

Find the equilibrium price and quantity.

Answer- $P^* = 2$; $Q^* = 3$

14. An individual has a budget of $120, which is to be spent on goods X and Y. Price of X and Y is given as $3 and $5 respectively.

(a). Draw a budget line using the budget line equation using all the different combinations of the two goods that can be bought with the given budget.

(b). What happens to the original budget line if the budget falls by 25%.

(c). What happens to the budget line if the price of X doubles.